

HUMAN- ROBOT INTERACTION

—— Lorenza, Rodrigo, Emanuel, Martin, Ana ——



CentraleSupélec

1

Team, context and objective

2

Operational flow

3

Technical challenges

4

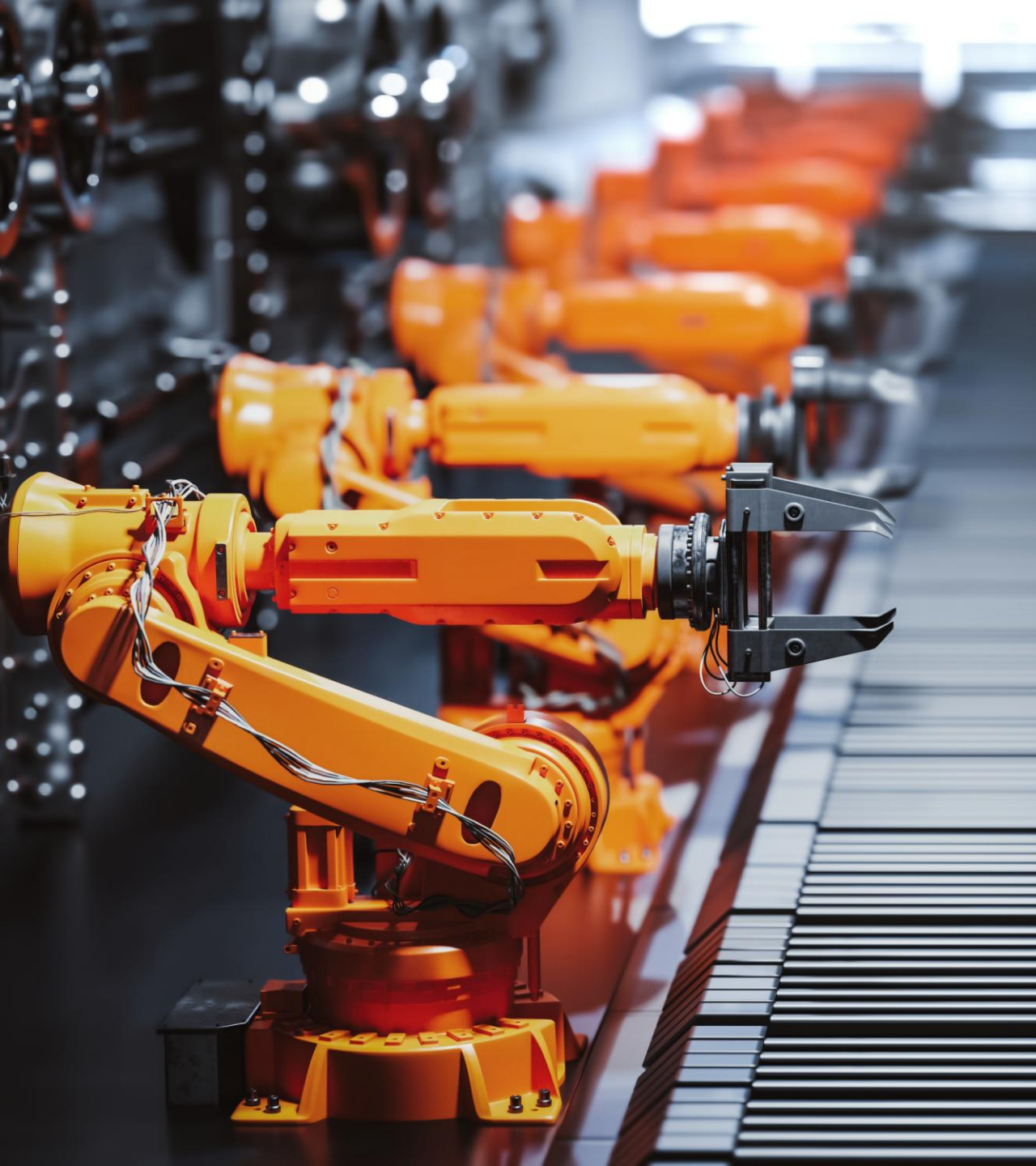
Achieved solution

5

Performance metrics

6

Closure -- (what can be improved)



1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

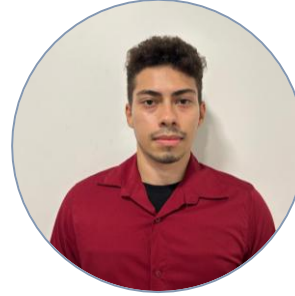
Performance metrics

6

Closure



Lorenza



Emanuel



Ana



Rodrigo



Martín



PRECISION

REPEATABILITY

PRODUCTIVITY and CONTINUITY

SAFETY

VERSATILITY

542,000

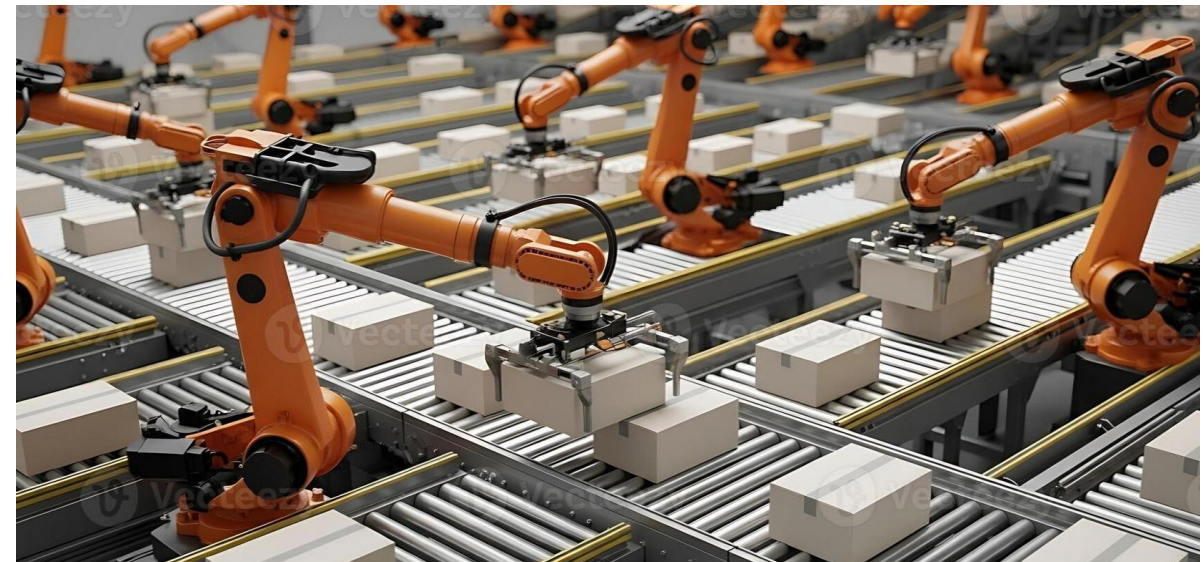
4.664 MILION UNIT

VISION

STRUCTURED INDUSTRIAL
ENVIROMENTS

PICK AND PLACE

SMART – INTERACTION
AND QUALITY CONTROL



PICK AND PLACE



1

VISION



2

INTERACTION

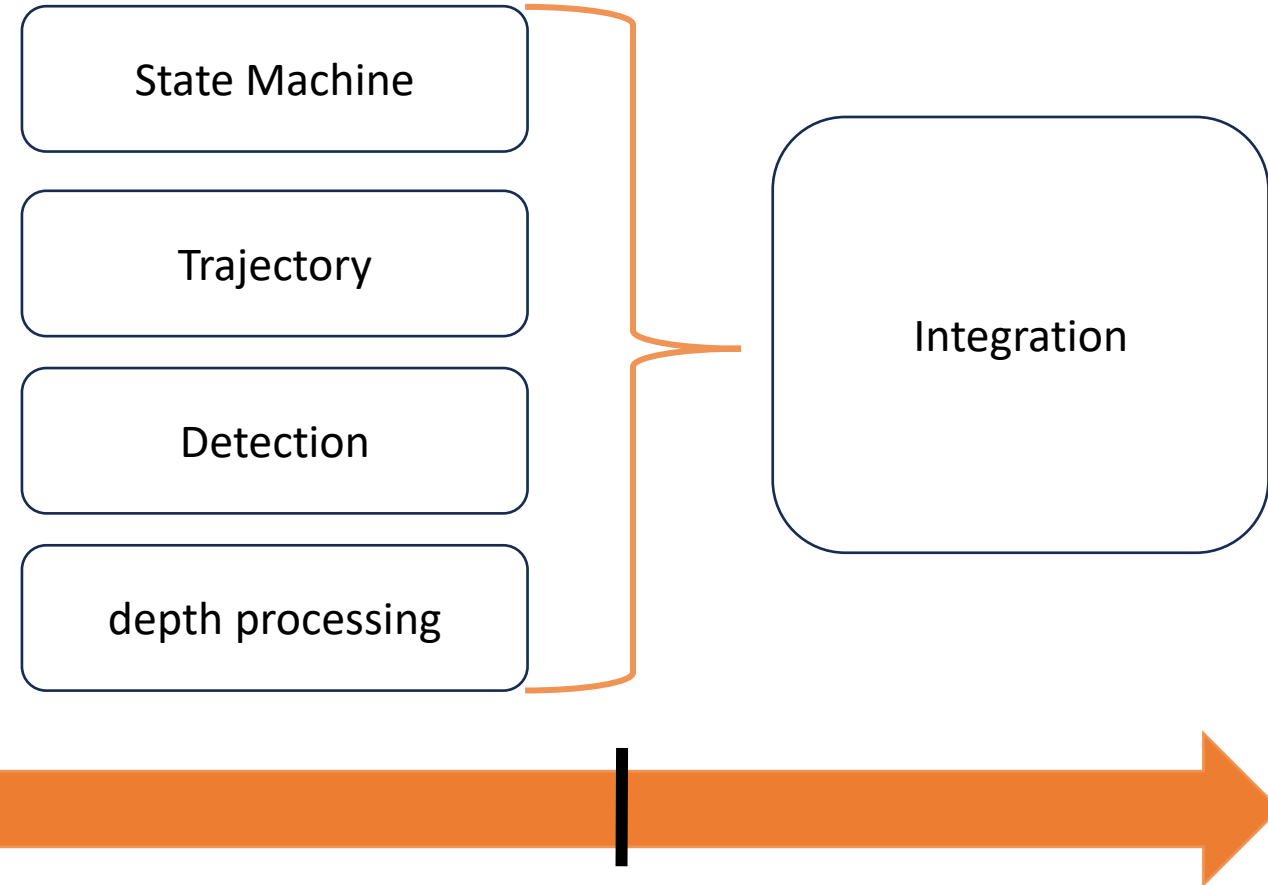


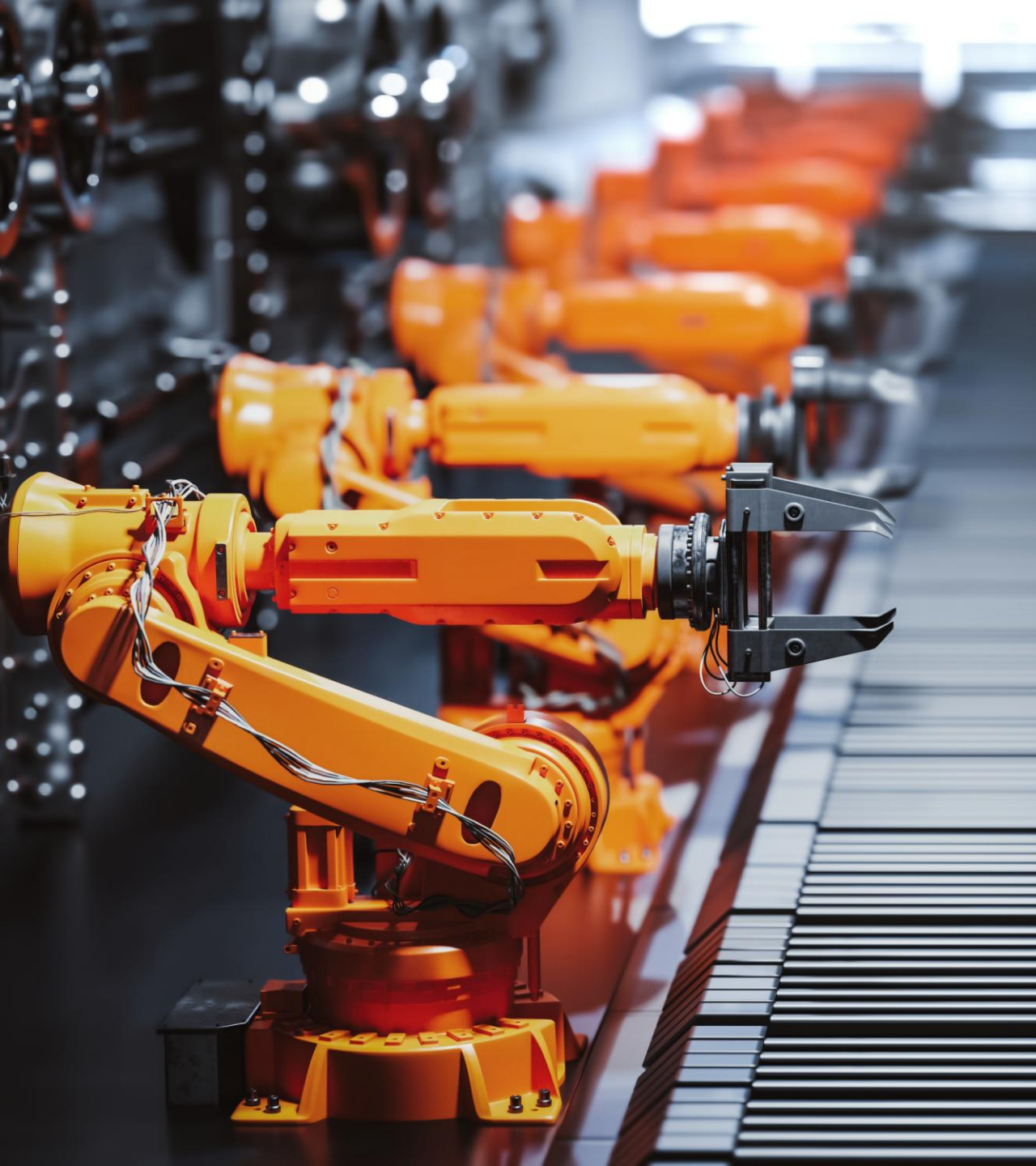
3

Course of action

State objectives and tasks

Divide task by area between the integrants





1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

Performance metrics

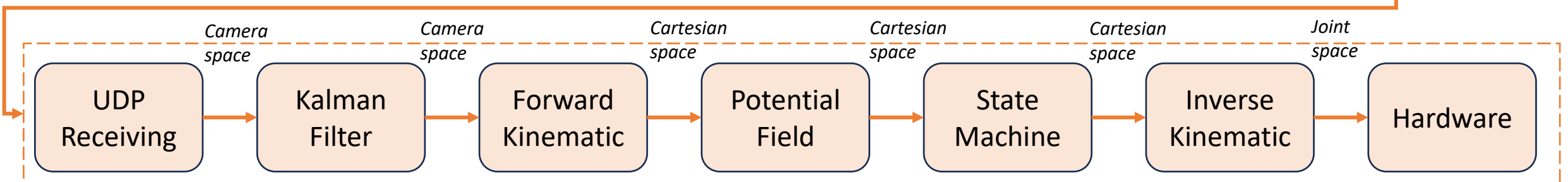
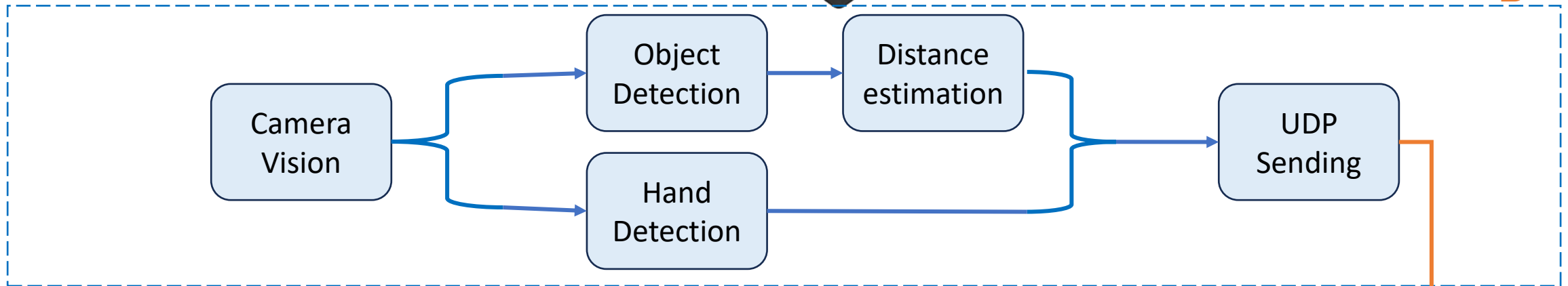
6

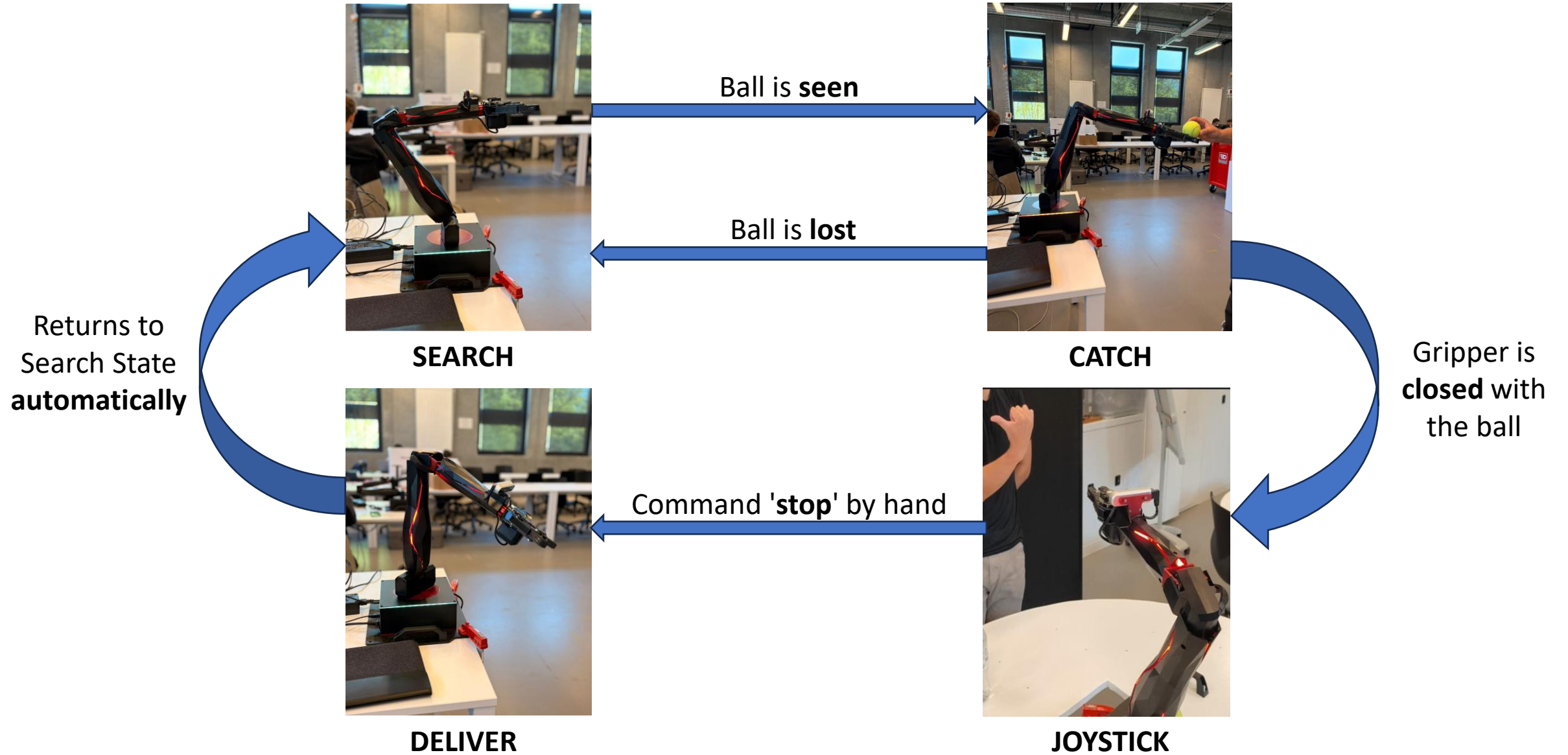
Closure

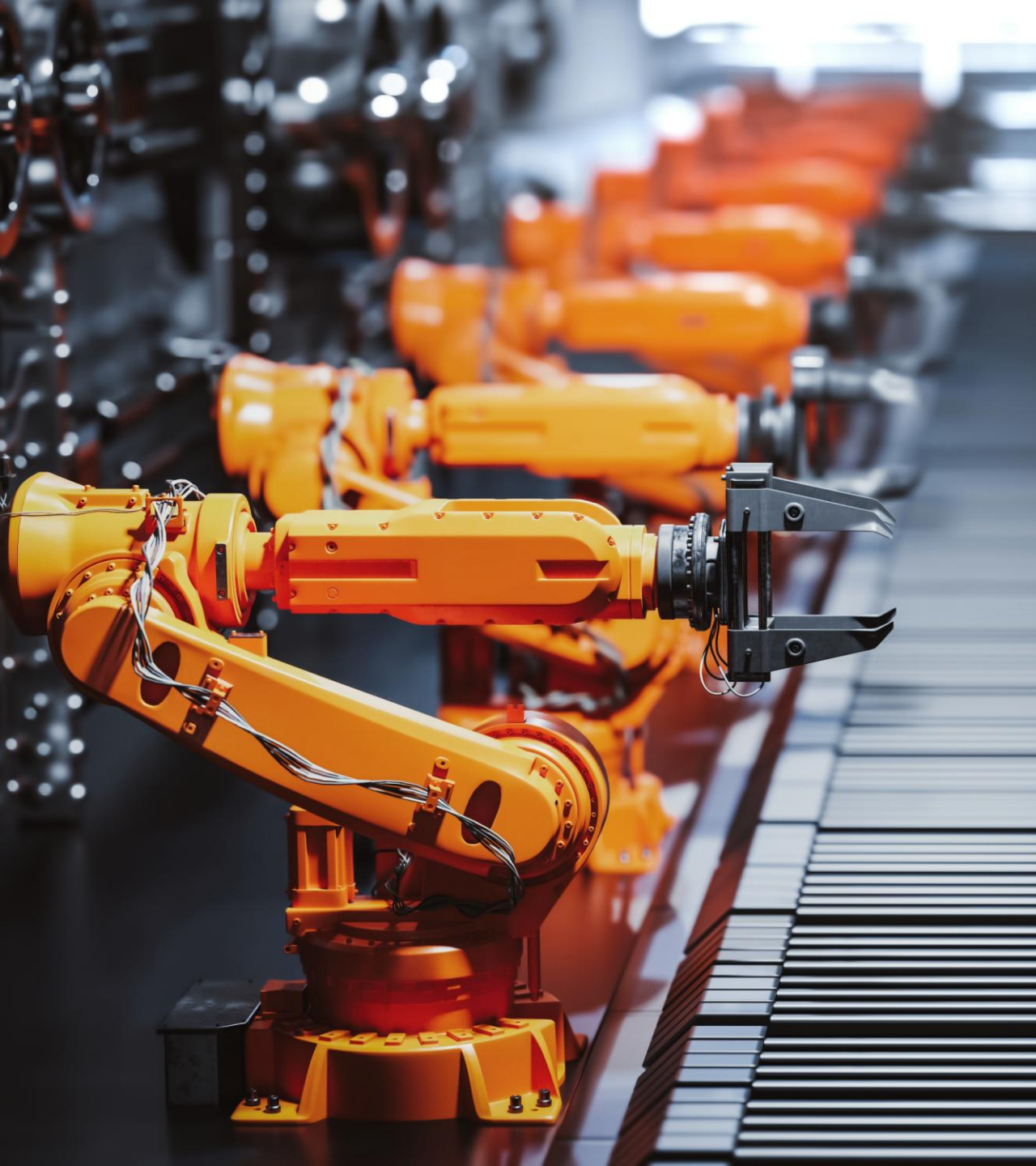
Vision part with Python



Trajectory calcul with MatLab







1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

Performance metrics

6

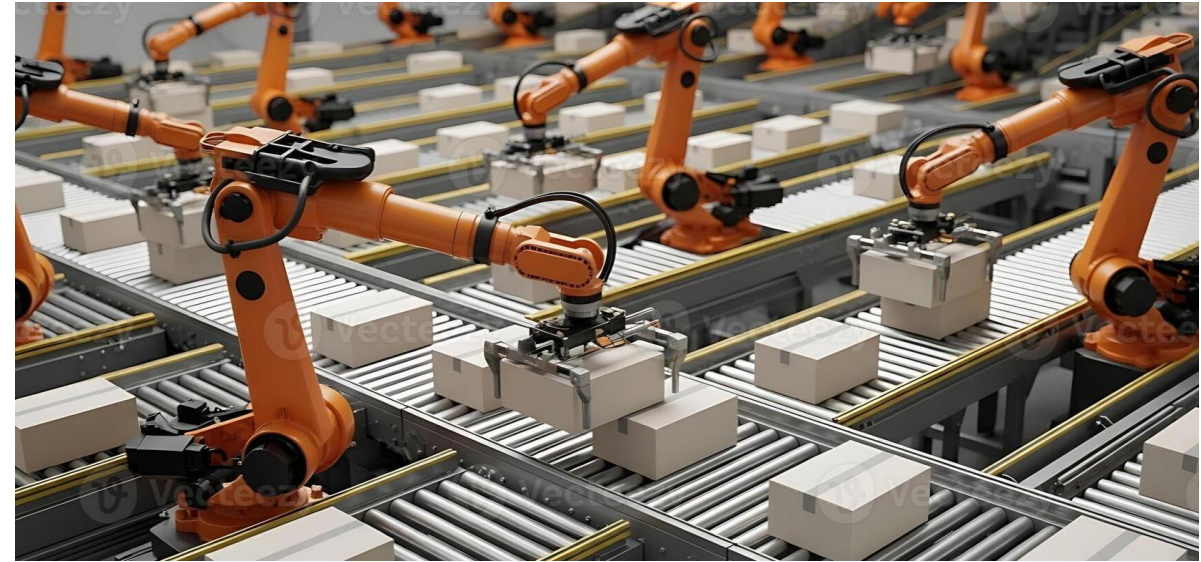
Closure

3 MAIN PROBLEMS

Logic between the different steps

Depth sensor not working below 30 cm

Fluctuation of the objective detection



3 MAIN SOLUTIONS

State Machine

Pinhole Camera Model

Kalman Filter

Bottle Detection Model & Gesture Detection

- Used YOLOv8 object detection model.
- Compared public datasets vs. custom dataset (300 images of the target bottle).
- Result: custom model with **96% accuracy**.



Hands Detector: 

- Camera captures hand gestures in real time.
- Each gesture → specific robot action (go right, go left, release, etc.).
- Result: clear, fast, reliable communication.
- Follow and release comands.

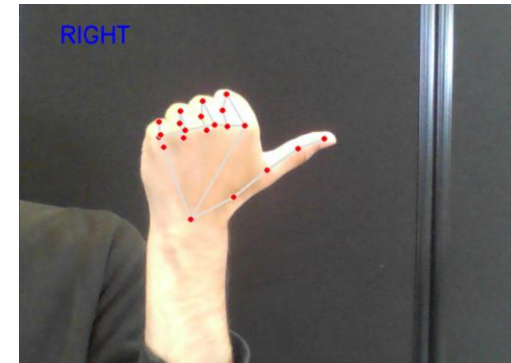
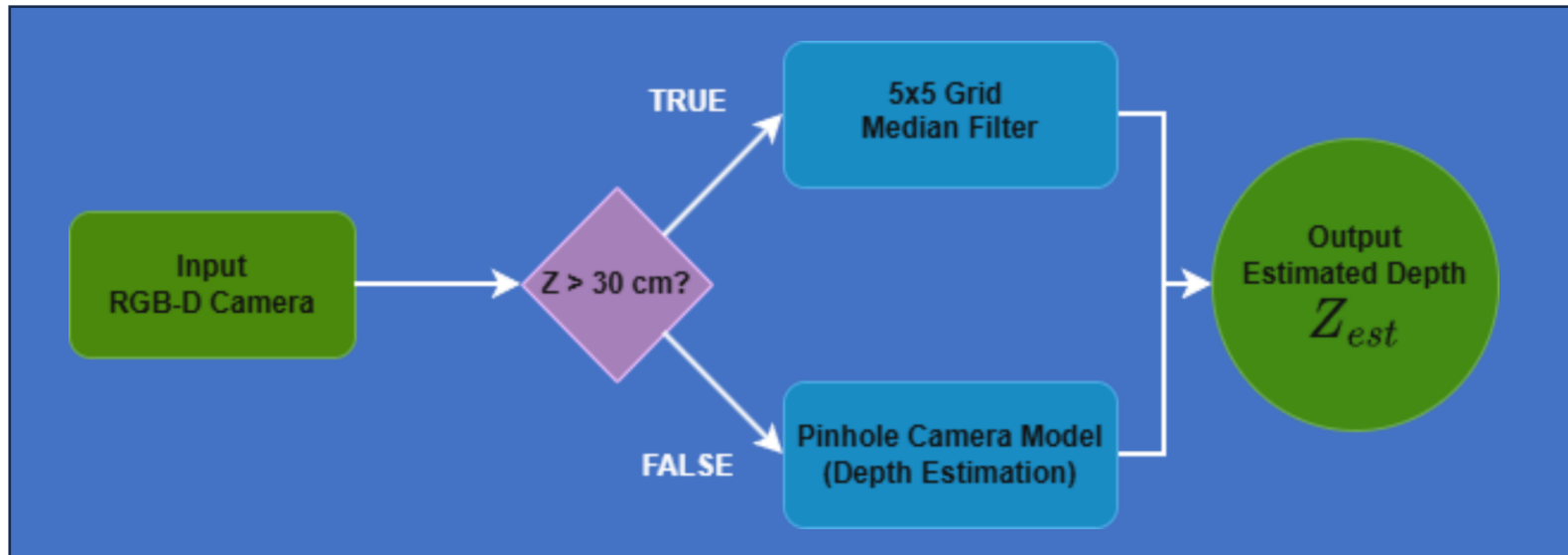


IMAGE & DEPTH PROCESSING

Problem: Intel RealSense D415 does not detect depth for distance $Z < 30\text{ cm}$

Solution:



Pinhole model equation:

$$Z_{est} = \frac{f_x \cdot D_{real}}{w_{pixel}}$$

f_x : Camera Focal Length (pixels)
 D_{real} : Physical Object Diameter (m)
 w_{pixel} : YOLO Bounding Box Width (pixels)

KALMAN FILTER

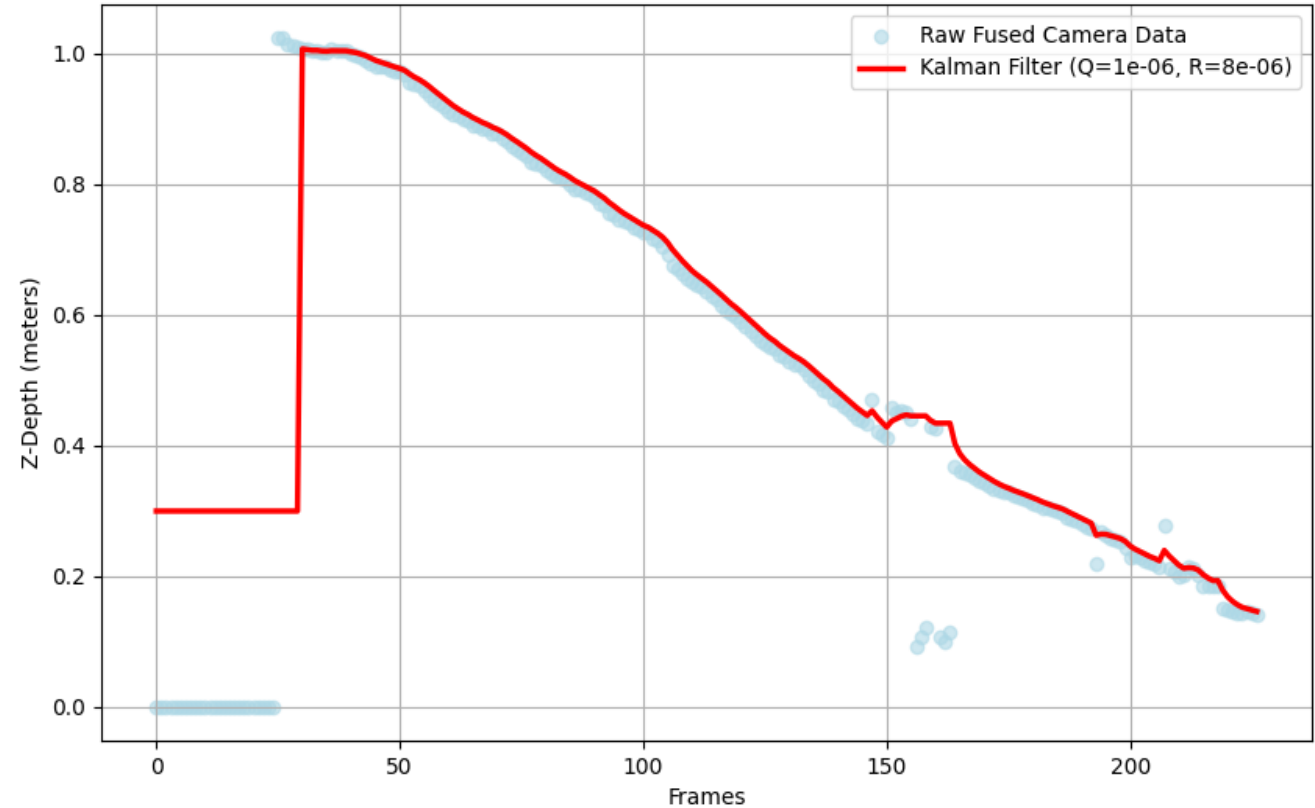
Implemented:

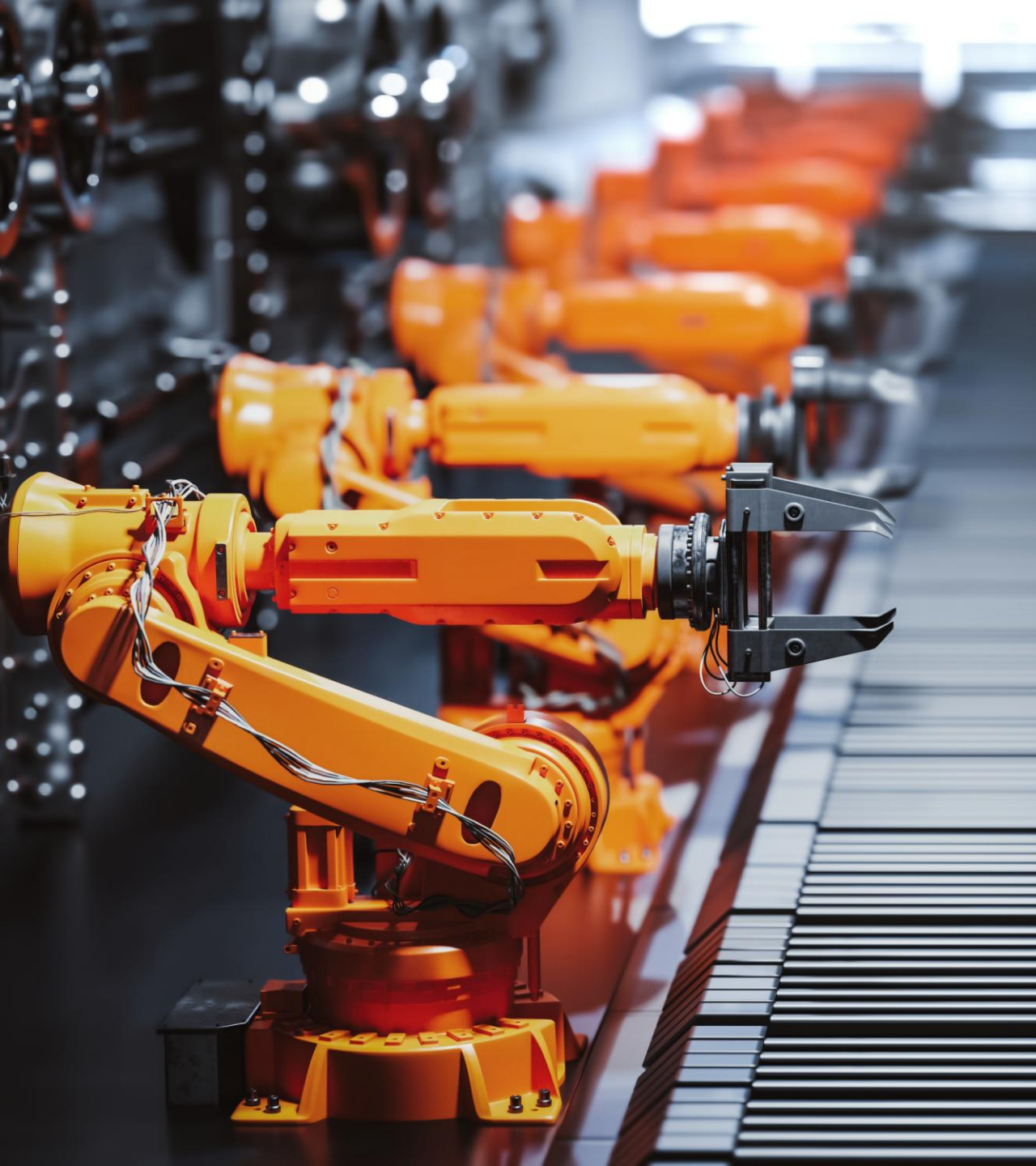
- Dual-Model Sensor Fusion
- Outlier Rejection Gate of $\pm 15cm(\Delta Z)$
- Total rejection of instant zero drops

Parameters:

- Measurement Noise (R): $8 \times 10^{-6} m^2$
- Process Noise (Q): $1 \times 10^{-6} m^2$

Kalman Filter Tuning results





1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

Performance metrics

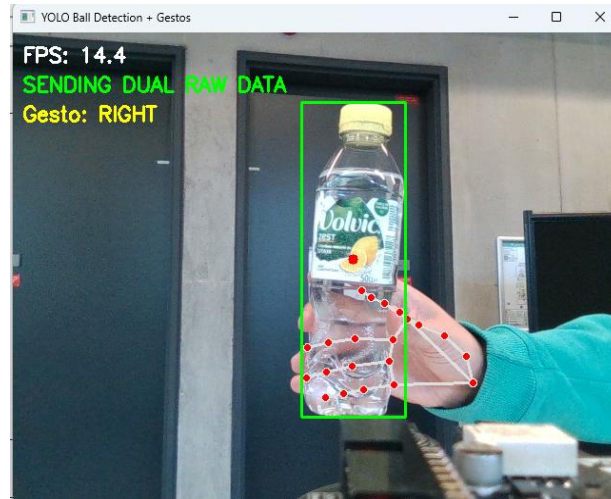
6

Closure



30cm

1. Solved the **short-distance** detection problem.



2. Trained a custom YOLOv8 model for bottles (**96%** accuracy).



3. Integrated MediaPipe Hands to recognize 3 gestures

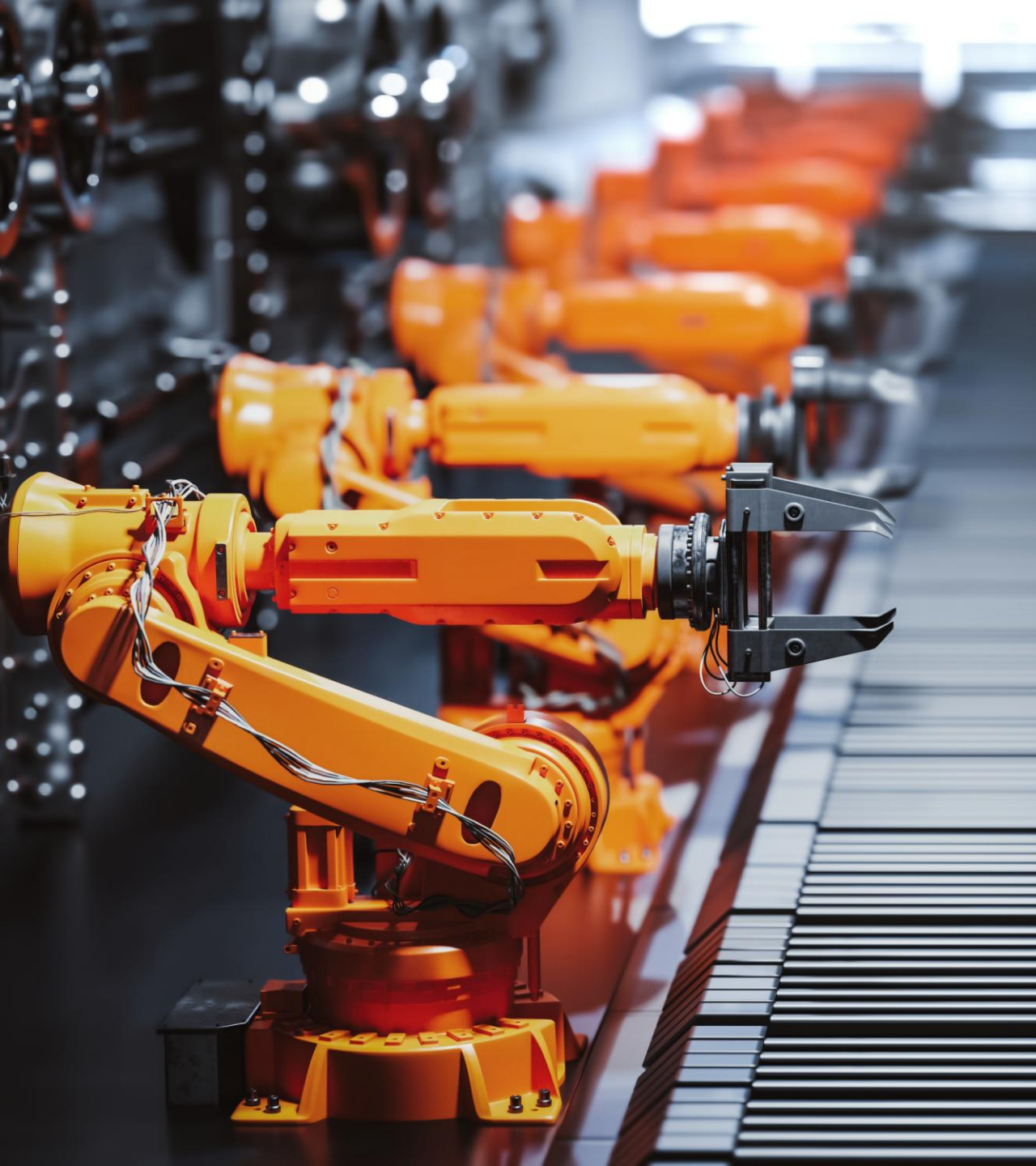


4. Designed a finite state machine to coordinate the task.

5. Realized a reliable, smooth trajectory, with singularity avoidance



6. Achieved **full bottle transfer** from one person to another



1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

Performance metrics

6

Closure

Detection model accuracy

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	640: 100%	14/14	1.0s/it	14.3s
50/50	2.21G	0.2725	0.2865	0.7967	14	mAP50	mAP50-95): 100%	2/2	3.5it/s	0.6s
	Class	Images	Instances	Box(P	R	0.995	0.96			
	all	55	55	0.999	1					



Experimental Methodology

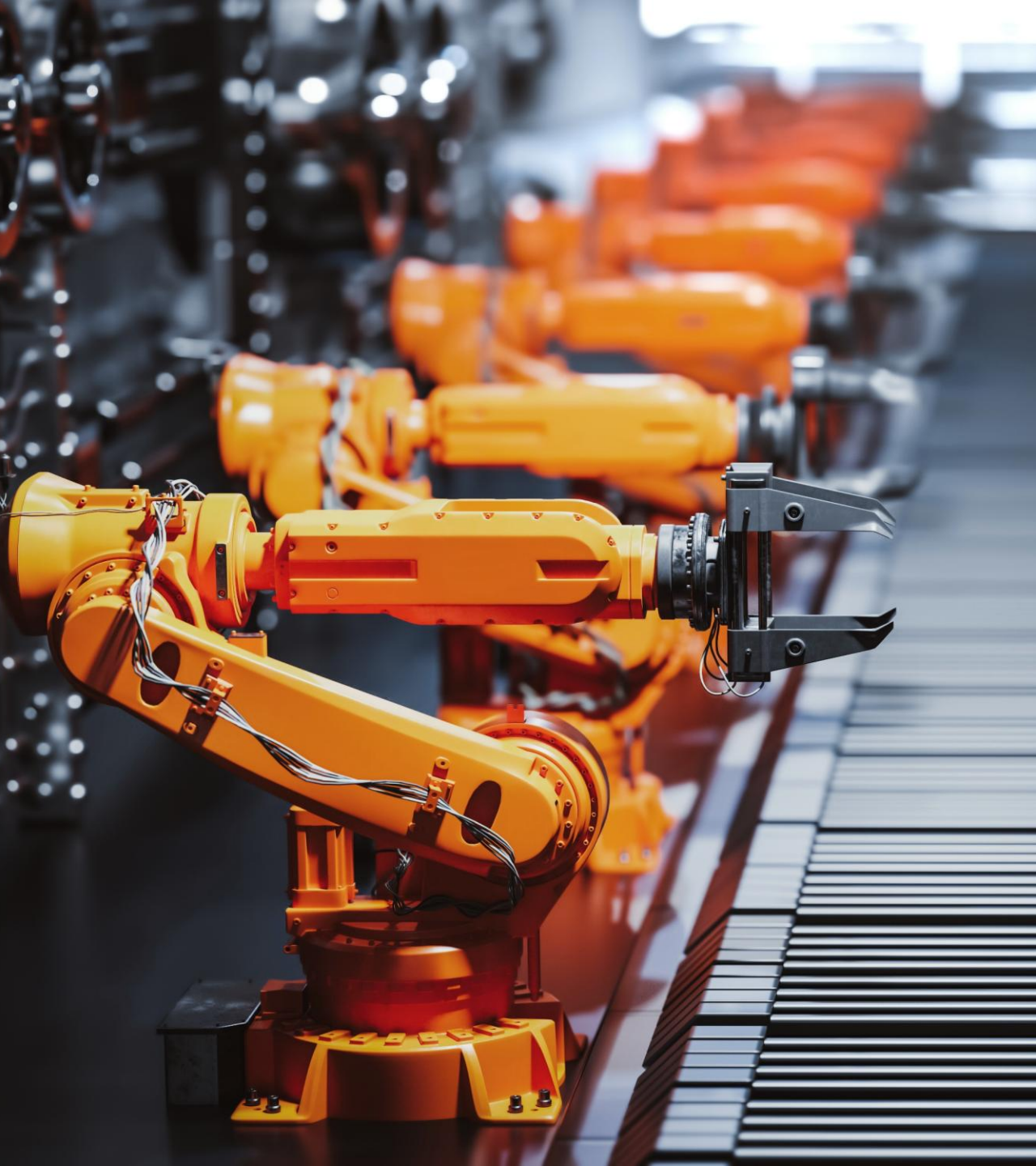
- 50 consecutive physical handover trials
- Bottle placed 50 cm in front of the QArm
- Bottle held strictly vertical

Quantitative Performance

- Successful grasps: 40/50 -> 80% Success rate
- Kinematic reliability: 100% (zero trajectory errors or singularities reached during movement)

Failure Analysis

- Primary failure: Premature gripper activation (100% of registered errors)
- Root cause: Visual noise anomalies persisted for more than 5 consecutive frames



1

Team, context and objective

2

Operational flow

3

Technical challenges

4

Achieved solution

5

Performance metrics

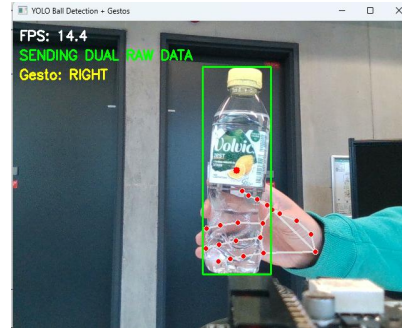
6

Closure

Project outcomes :

Perception

- Custom YOLO model (**96%** accuracy)
- **Gesture** recognition
- Short-range depth **estimation**



Control

- **Kalman** filtering
- Finite **State** Machine
- **Potential** Field navigation

Demonstration

- **Bottle** detection
- Autonomous **grasping**
- **Smooth** switch of states



Objectives for the future :

- **Wrist** orientation control
- False positive **rejection**
- **Extended** gesture vocabulary
- Grasp **verification** feedback
- Obstacle **avoidance**
- Dynamic target **prediction**
- Trajectory **smoothing**
- **Multi-object** interaction

The background image shows a factory floor with several yellow industrial robotic arms. Two arms in the foreground are positioned symmetrically, facing each other. They have black grippers at the end. The floor is a metal grating. In the background, other robotic arms and factory equipment are visible, slightly out of focus. A semi-transparent dark red rectangle is centered over the image, containing the text "THANK YOU FOR YOUR ATTENTIONS" in white, bold, sans-serif capital letters. The text is flanked by two short white horizontal lines.

THANK YOU FOR YOUR ATTENTIONS

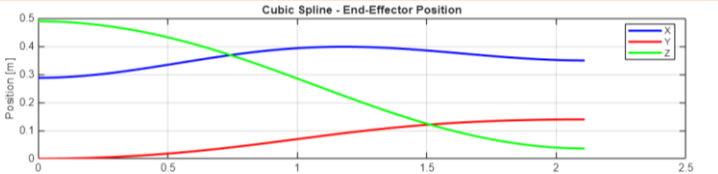
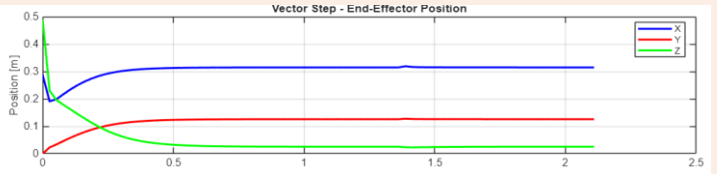
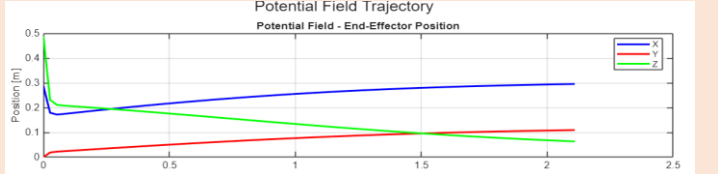


Search Mode

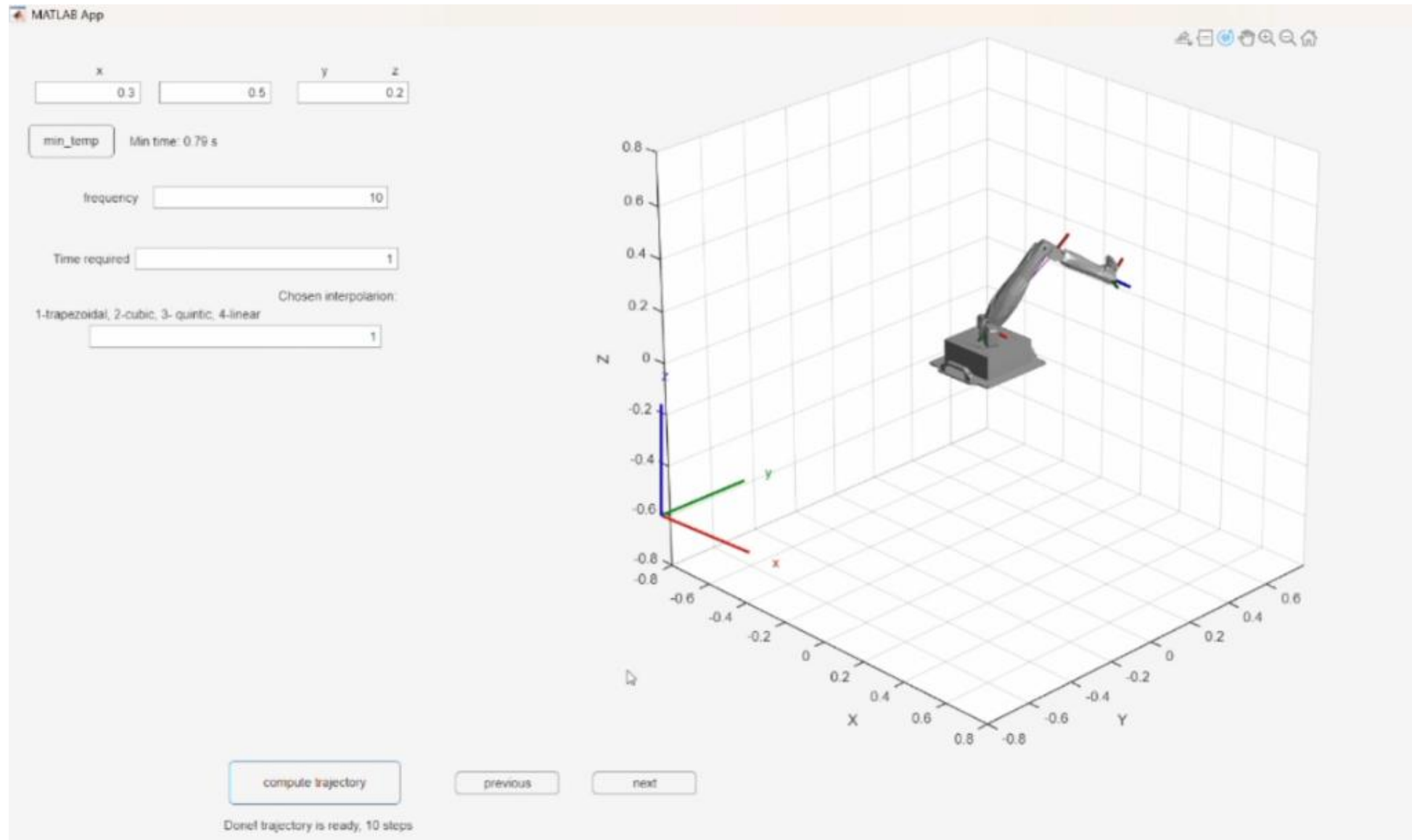


Joystick Mode

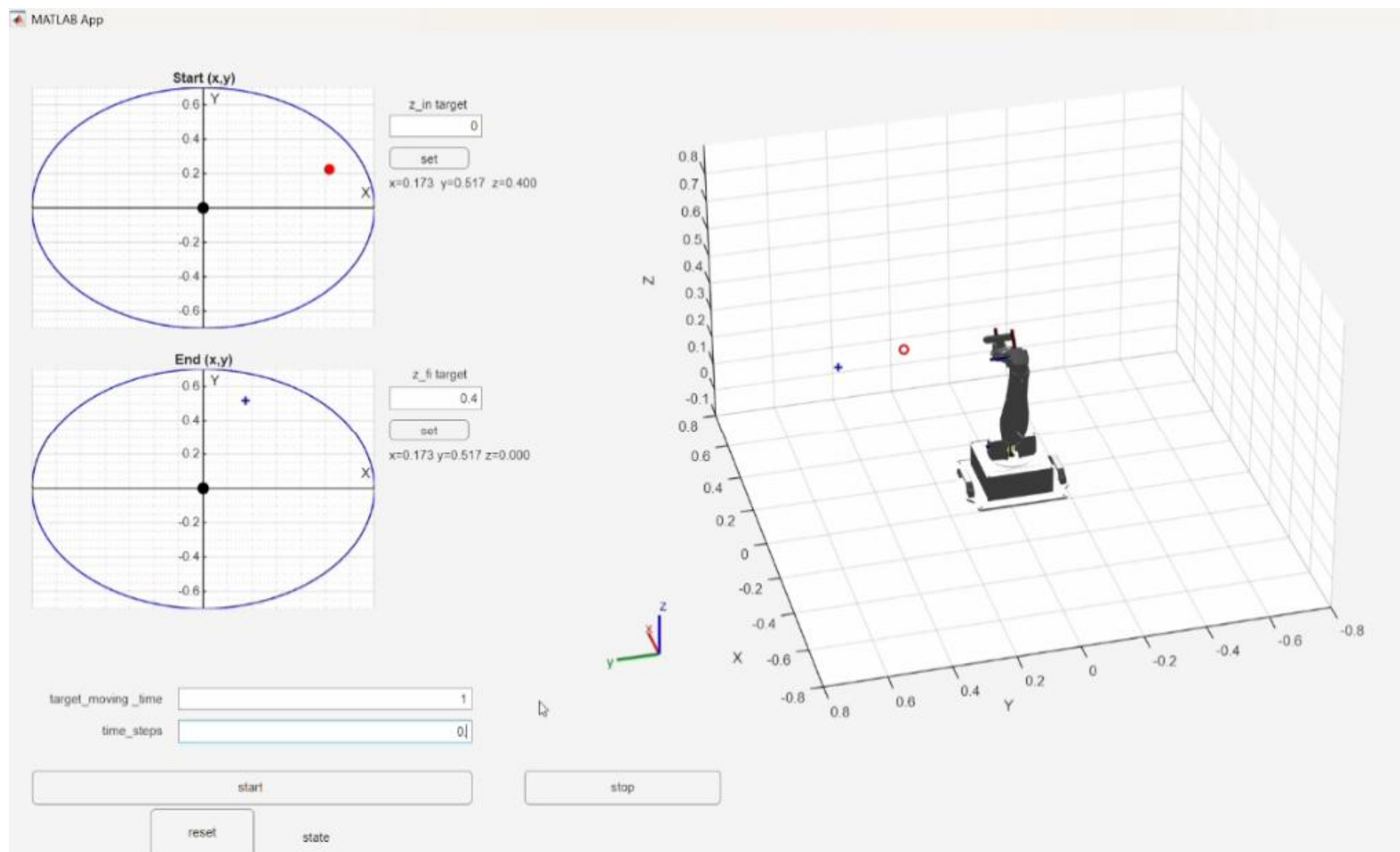
Trajectory Choice

Mean XYZ	Cubic Spline	Potential Field
Initially used for object tracking	Used in the original Pick-and-Place application	Final solution selected
• Very simple implementation • Low computational cost	• Smooth and continuous trajectories • Predictable motion	• Real-time trajectory generation • No predefined path required
• Tracking lag due to averaging • Limited trajectory control	• Requires trajectory replanning each time the target moves • Less suitable for dynamic environments	• Can generate non-smooth motions • Sensitive to parameter tuning
		

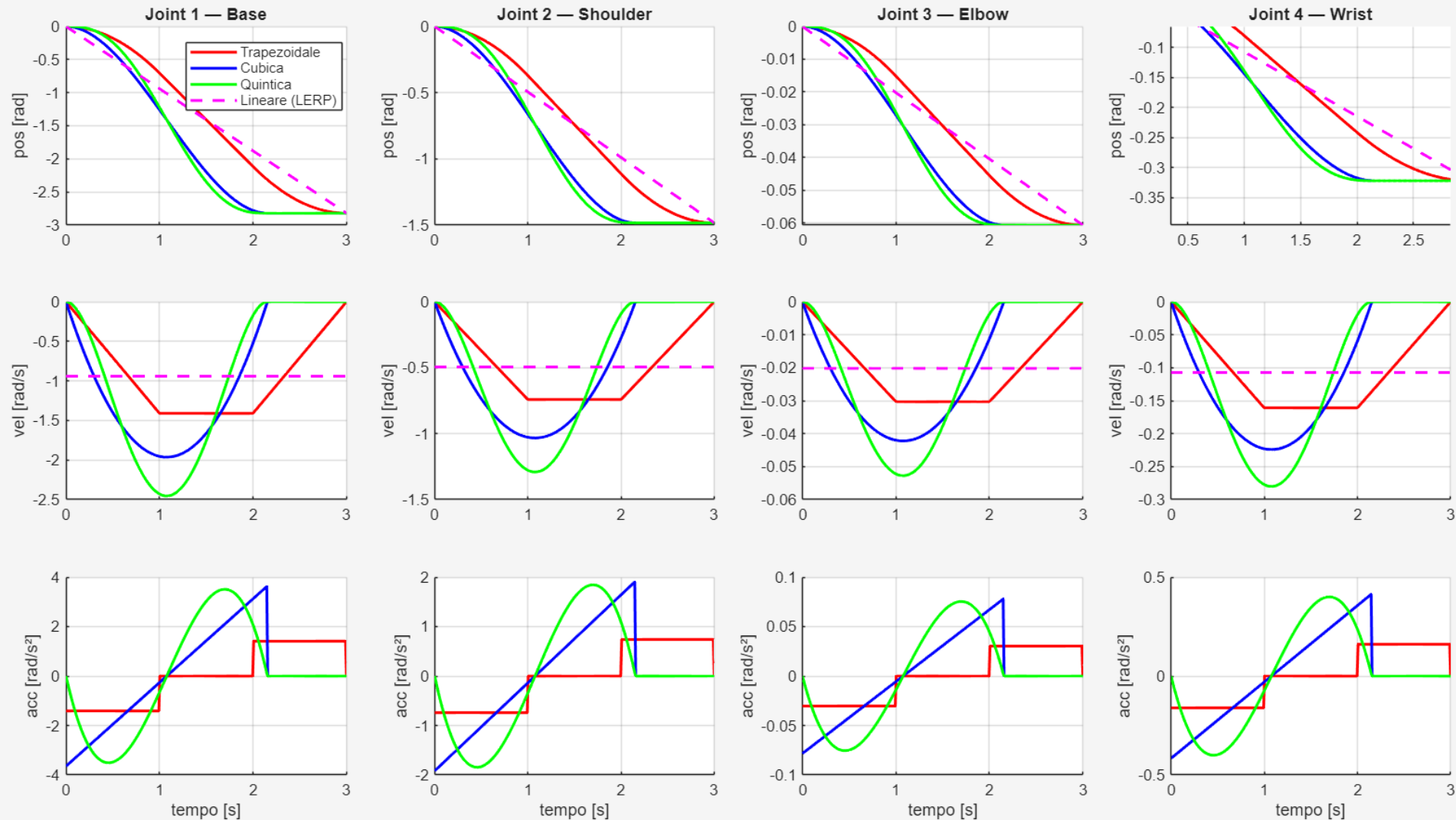
SPLINE INTERPOLATION



POTENTIAL FIELD



Confronto Traiettorie — QArm: Trapezoidale / Cubica / Quintica / Lineare



JOINT: TRAPEZOIDAL, CUBIC, QUINTIC, LINEAR

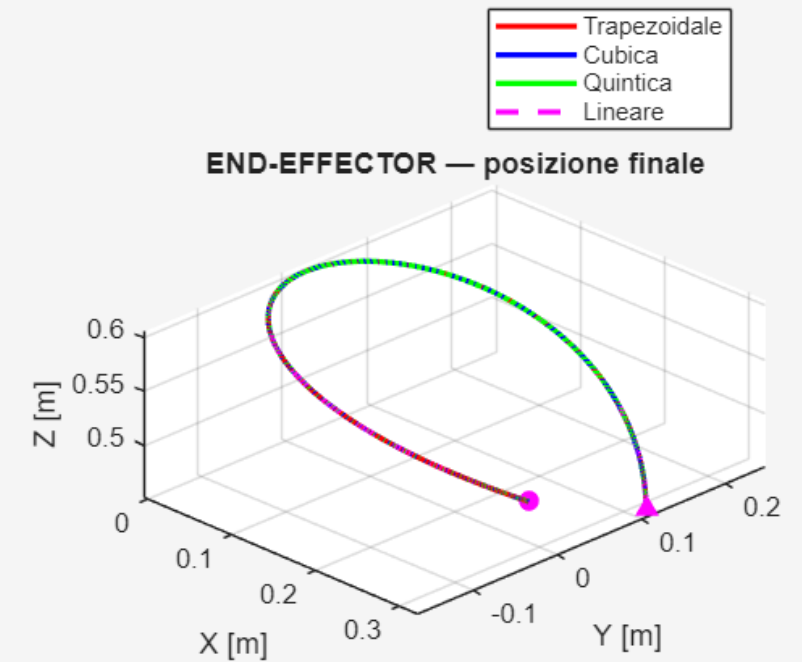
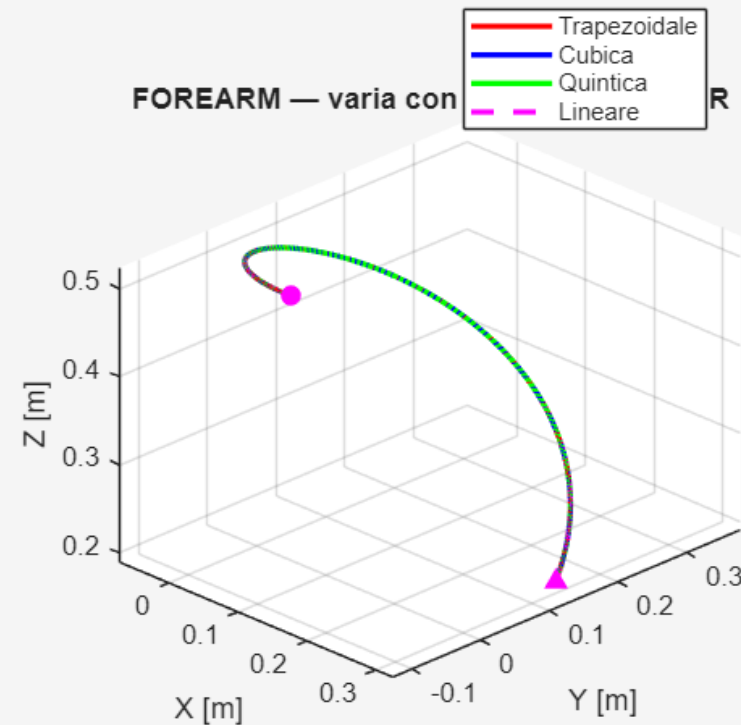
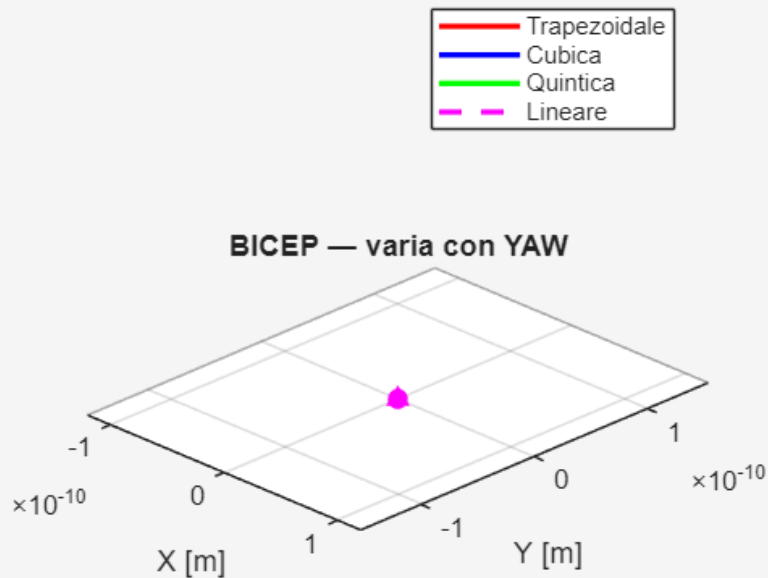
```
dqMax = deg2rad([90 90 90 90]); %90 degree/ s
```

```
dq = abs(q_end - q0); % 1x4 one for joint
```

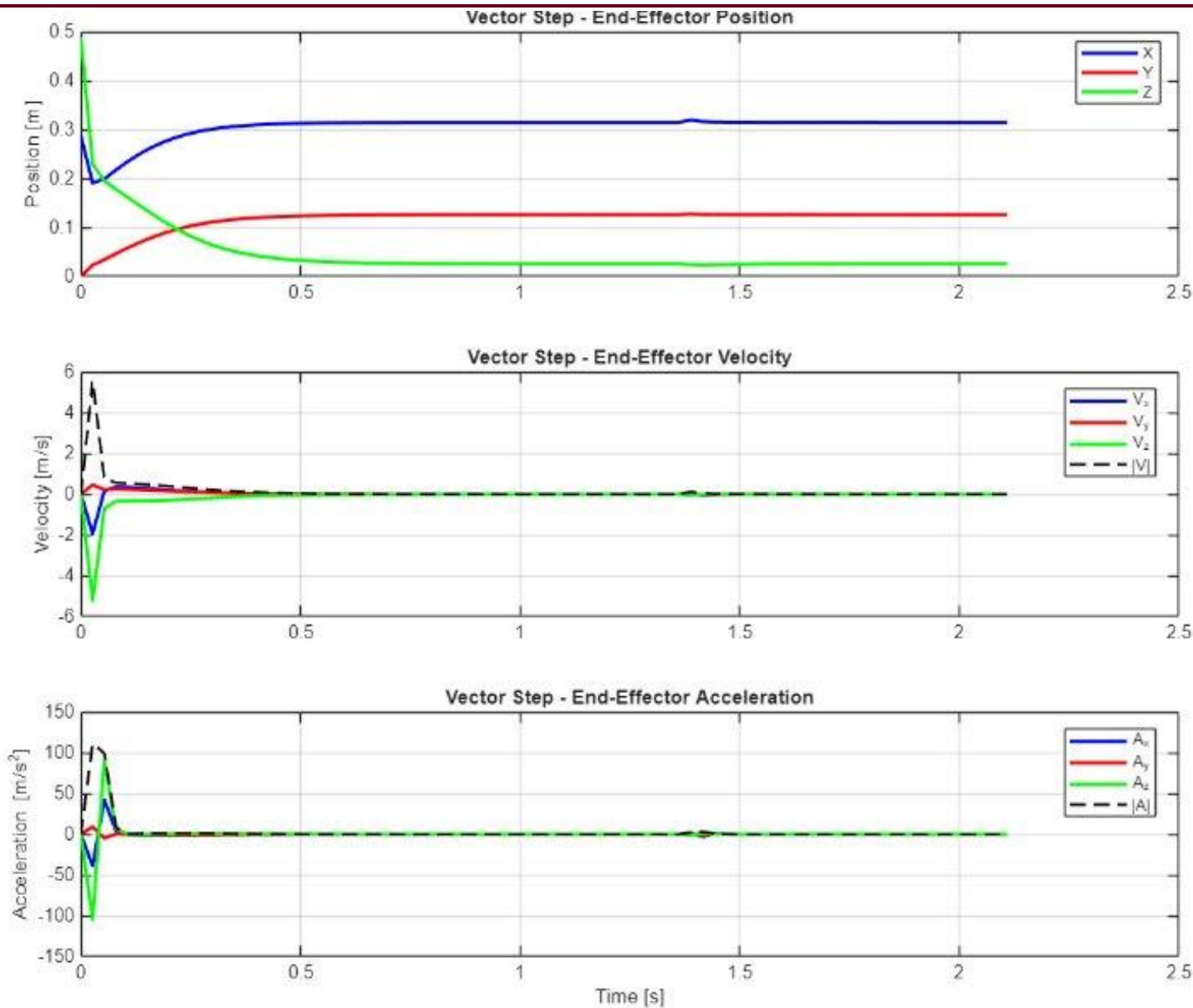
```
T_min = max(dq ./ dqMax); % 2) T_min required to respect the constraints
```

```
safetyFactor = 1.2; scaling by a safety factor
```

```
T = safetyFactor * T_min;
```

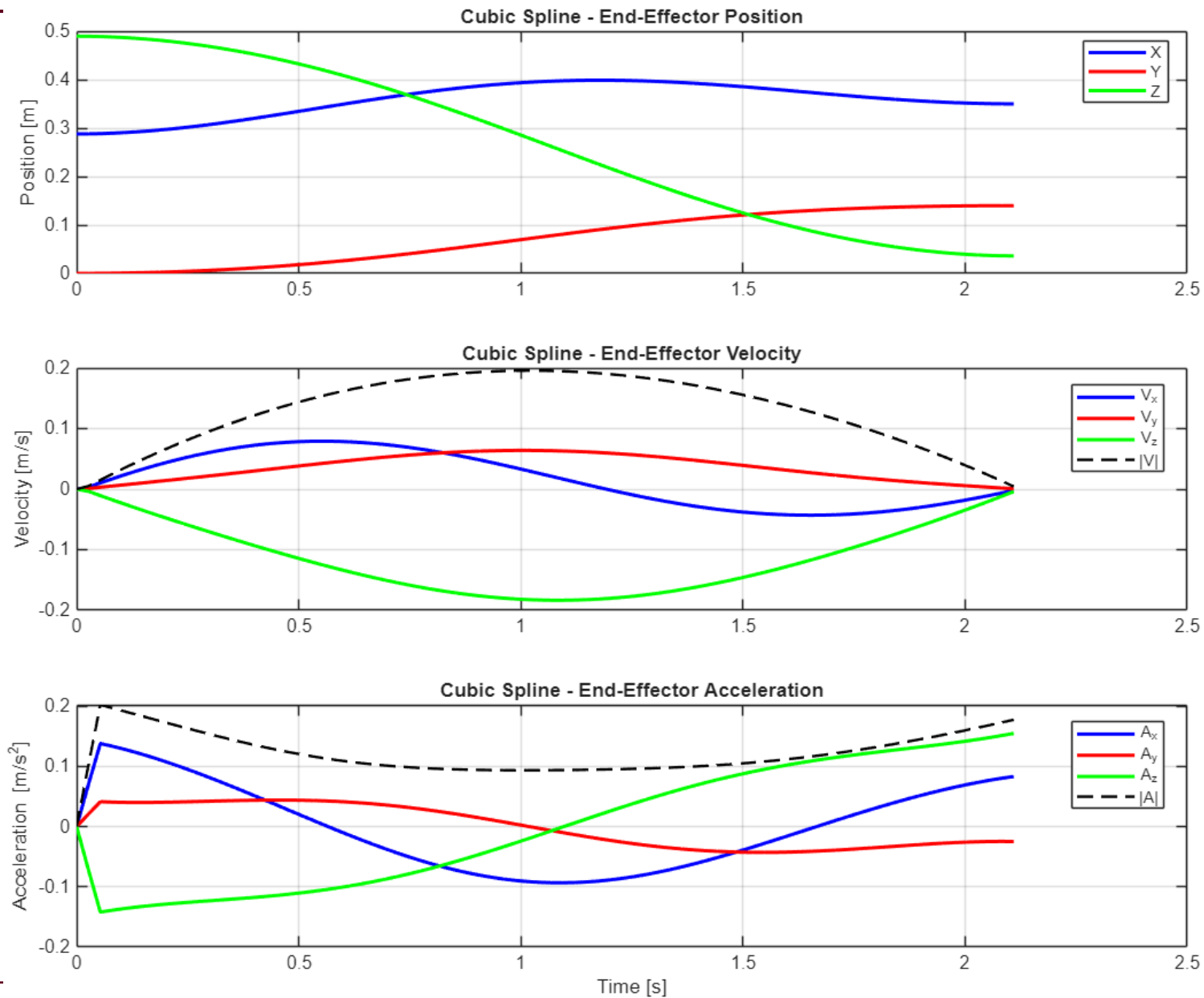


MEAN XYZ



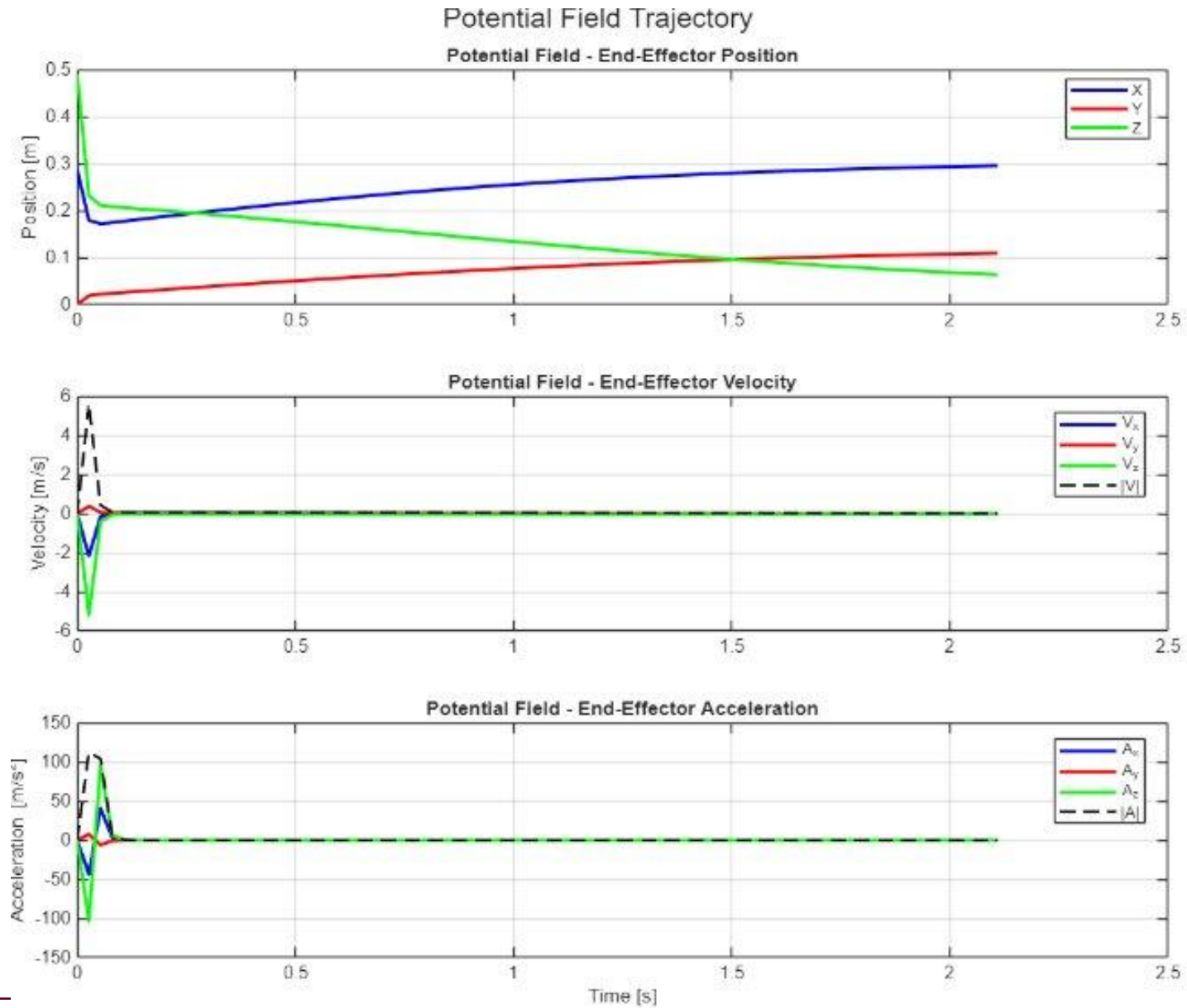
`p_target = [0.50; 0.20; 0.10];`

CUBIC SPLINE



$p_target = [0.50; 0.20; 0.10];$

POTENTIAL FIELD



$p_target = [0.50; 0.20; 0.10];$